



AQLI WhatsApp Chatbot

Professional User and Administrator Manual

Country and state/province AQLI analysis in WhatsApp

PM2.5 values | Life-expectancy impact | Rankings | Trends | Comparisons

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WhatsApp: +91 96547 01203

Admin portal: <https://aqli-whatsapp-chatbot.vercel.app>

Prepared for AQLI chatbot users, analysts, and administrators

Document guide

This manual explains how to use the AQLI WhatsApp chatbot, how to interpret its answers, and how administrators maintain conversations and knowledge sources. It reflects the production behavior verified on 4 July 2026.

Current data boundary: The structured database contains country-level (GADM0) and state/province-level (GADM1) data through 2024. District-level (GADM2) data is not currently active.

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About the chatbot

The AQLI WhatsApp chatbot is a conversational interface for the Air Quality Life Index. It lets a user ask ordinary-language questions about long-term PM2.5 pollution and estimated life-expectancy impact without opening a spreadsheet, report, or analytical dashboard.

What AQLI means

AQLI means Air Quality Life Index. It translates long-term exposure to fine particulate matter (PM2.5) into an estimated effect on life expectancy. The core relationship used by the chatbot is approximately 0.98 years of life-expectancy loss for each 10 µg/m³ increase in long-term PM2.5 above the selected benchmark.

```
Life loss per person = max(PM2.5 - benchmark, 0) x 0.098
```

When the uploaded CSV already contains pre-calculated life-loss values, the chatbot uses those fields directly instead of recalculating them.

Primary audiences

- Policy and program teams that need quick, referenced AQLI facts.
- Researchers and analysts checking values, trends, rankings, and comparisons.
- Communications teams, journalists, educators, and public users seeking short explanations.
- Administrators who upload data and reports, monitor conversations, or take over a chat.

Quick start for WhatsApp users

Start a conversation

1. Open WhatsApp and start a chat with +91 96547 01203.
2. Send a greeting such as Hi, Hello, or Good morning.
3. Ask one focused question using a place, metric, and optional year or period.
4. Read the answer, data-coverage notice, and reference line.
5. Ask a follow-up or send reset to begin a new topic.

Fastest first test: Send: What is India's PM2.5 in 2024? Then ask: Compare it with China.

Quick-reference examples

Need	Example question
Latest value	What is India's latest PM2.5?
State ranking	Give top 5 most polluted states in India.
Threshold list	List countries with PM2.5 above 30 in 2024.
Comparison	Compare India and China from 2010 to 2020.
Mixed level	Compare India and Uttar Pradesh for the last decade.
Trend	Show the PM2.5 trend for Bihar.
Benchmark	What is Jharkhand's life loss under the national standard?
Methodology	How does AQLI calculate life-expectancy loss?

How to write a good question

The chatbot handles natural wording, common misspellings, shuffled word order, and many comparison phrases. Accuracy is still highest when the question contains the following elements.

Element	What to provide	Examples
Geography	Country or state/province	India; Bihar; Uttar Pradesh
Metric	PM2.5, life loss, population, standard	PM2.5; life years lost
Time	A year, range, or relative period	2024; 2010-2020; last decade
Benchmark	WHO, national, or custom target	WHO; national std; target 15 µg/m ³
Operation	Value, rank, compare, trend, or filter	top 5; compare; above 30

Recommended question pattern

[Action] + [place/level] + [metric] + [year/period] + [benchmark]

Example: Compare Bihar and Uttar Pradesh life loss from 2014 to 2024 using the national standard.

When the question is ambiguous

The chatbot asks one short clarification instead of guessing. For example, a request for top states without a country prompts for the country. A request for top 10 states every year may ask whether the user wants a period average or the single annual leader.

Bot: Do you want (1) the top 10 states averaged across the period, or (2) the single most polluted state for each year?
User: 2

Supported question types

Direct values

Use direct-value questions for one location and one year. If no year is supplied, the latest available year is used. The current latest year is 2024.

- What is the PM2.5 level in Bihar?
- What was India's PM2.5 in 2020?
- What is the population of Nepal in the latest year?
- What is life-expectancy loss in Jharkhand under the national standard?

Rankings

Rankings may be requested for countries or states/provinces. State rankings normally require a country unless the question explicitly asks for a worldwide state ranking.

- Top 10 countries by PM2.5 in 2024.
- Give top five most polluted states in Nepal.
- Which state has the highest life loss in India?
- Which countries improved the most over the last decade?

Ranking rule: PM2.5 and life loss are proportional for the same benchmark, so the bot normally shows both rather than asking which one to rank.

Threshold filters

Threshold questions return matching places with units. The parser distinguishes strict, inclusive, and exact boundaries.

Meaning	Recognized examples	Behavior
Strictly above	more than, greater than, above, over, exceeds, >	Value must be greater than the threshold
At or above	at least, no less than, >=	Threshold value is included
Strictly below	less than, lower than, below, under, <	Value must be lower than the threshold
At or below	at most, no more than, <=	Threshold value is included
Equal	equal to, exactly, same as, =	Only exact matches are returned

Examples: List states in India with PM2.5 below 30. List countries with national std more than 30. How many countries have PM2.5 at least 35?

Comparisons

The chatbot compares two countries, two states, or a country and a state. Country values are taken from GADM0; state/province values are taken from GADM1. Mixed-level answers include a level note so the user understands that the geographic units differ.

```
Compare India and China in 2024.  
Compare Bihar and Uttar Pradesh for the last decade.  
Compare India and Uttar Pradesh from 2014 to 2024.
```

A period comparison shows the start value, end value, change, direction, final-year difference, and life-loss change for both places.

Trends and time periods

Trend questions show the first year, last year, PM2.5 change, life-loss change, and whether long-term exposure improved or worsened. Relative periods end at the latest available year. Use an explicit range when exact endpoints matter.

- Show the trend for Madhesh.
- How did Delhi change from 2010 to 2024?
- Compare India and China over the past ten years.
- What was the highest PM2.5 in Uttar Pradesh during the last 20 years?

Annual leaders and period averages

For multi-year rankings, the chatbot can calculate a period average or return the single leader for each year. If a top-N annual request could mean either, it asks the user to choose.

```
Top 10 most polluted states in India between 2010 and 2024.  
Give the most polluted state in India for each year from 2010 to 2024.
```

Benchmarks

Benchmark	How to request it	Data behavior
WHO guideline	WHO, WHO guideline	Default when no benchmark is specified; uses llpp_who when available
National standard	national standard, national std, nat std, national limit	Uses llpp_nat/llpp_national when available
Custom target	target 10 µg/m ³	Calculates $\max(\text{PM}_{2.5} - \text{target}, 0) \times 0.098$

Total life-years calculation

When requested, the chatbot calculates an aggregate using the user-defined convention below.

Total person-years lost = life loss per person x population

Interpret carefully: This aggregate is a calculated communication measure. It should not be presented as an official individual prediction or medical diagnosis.

Reports and methodology

Use uploaded annual reports and methodology documents for definitions, policy context, interpretation, limitations, and report findings. Numeric rankings and comparisons continue to use the structured CSV data first.

- What is AQLI?
- Why does AQLI use PM_{2.5}?
- How is life-expectancy loss calculated?
- What are the limitations of AQLI?
- Summarize the annual report's main findings.

Conversation memory and reset

Follow-up questions

The chatbot uses recent messages from the same conversation to interpret short follow-ups. It can retain the selected place, level, year, benchmark, comparison subject, or requested operation.

```
User: Tell me about Uttar Pradesh.  
User: Show its trend.  
User: Compare it with Bihar.  
User: Now use the national standard.
```

Start a clean topic

Send the single word reset when previous context is influencing a new question.

```
reset
```

Important: reset clears the active conversational context used for future answers. It does not delete stored conversation messages from Supabase.

When not to rely on memory

- After changing from a country question to an unrelated state question.
- After a long sequence containing several benchmarks or time periods.
- When a reply mentions a place different from the one intended.
- Before formal testing, demonstrations, or screenshots.

Understanding answers and references

Standard answer structure

1. Answer first: the value, ranking, trend, or comparison requested.
2. Short interpretation: included where it helps explain direction or significance.
3. Suggested next question: a relevant follow-up, when available.
4. Data coverage notice: country and state/province data only.
5. Reference line: the source level, year/range, benchmark, and conversation context when used.

Reference labels

Reference	Meaning
GADM0 CSV	Country-level structured data
GADM1 CSV	State/province-level structured data
GADM0 CSV + GADM1 CSV	Mixed country-to-state comparison using exact source levels
AQLI Annual Report	Narrative findings, context, and interpretation
AQLI Methodology document	Definitions, conversion relationship, assumptions, and limitations
Previous conversation context + ...	The current answer used a recent conversation selection
Reference checked	The bot checked the source but could not support the requested answer

Units

- PM2.5 concentration: $\mu\text{g}/\text{m}^3$.
- Life loss: years per person unless explicitly described as total person-years.
- Population: people.
- Change: $\mu\text{g}/\text{m}^3$ or years, with direction stated separately.

Data coverage, methodology, and limitations

Current structured data coverage

Level	Status	Use
GADM0 - Country	Active	Country values, rankings, filters, trends, and comparisons
GADM1 - State/province	Active	State/province values, rankings, filters, trends, and comparisons
GADM2 - District	Not currently active	District questions return a clear coverage limitation

Important limitations

- **Not live AQI:** The data represents annual long-term PM2.5, not current hourly air quality or an emergency alert.
- **Data freshness:** The current CSV series runs through 2024. New annual data must be uploaded before the bot can use it.
- **Grounded scope:** The chatbot does not invent unsupported values. Missing information produces a clear unavailable-data response.
- **Population estimates:** AQLI values describe estimated population-level effects, not an individual's medical outcome.
- **Geographic comparison:** Country-to-state comparisons are allowed but are explicitly labeled as different geographic levels.
- **Language:** The production parser is optimized for English. Other languages have not been fully validated.
- **Message format:** The webhook currently processes text messages only. Images, voice notes, documents, and other media are not answered automatically.
- **Platform dependency:** Replies depend on Meta WhatsApp Cloud API, Vercel, Supabase, and configured model/embedding services.
- **Conversation retention:** reset does not erase database records. Retention and deletion must be handled administratively.

User-facing disclaimer: Every automated reply states that current structured data coverage is country and state/province level only and that district-level data is not currently available.

Troubleshooting for users

Symptom	Likely cause	What to do
No reply	Webhook/platform issue, human mode, or unsupported media message	Send a text message. Wait briefly. Ask an administrator to check conversation mode and production logs.
Wrong place or topic	Old conversation context	Send reset, then ask the complete question again.
Bot asks for a country	A state ranking did not include its country	Reply with the country name, such as India or Nepal.
District unavailable	No active GADM2 dataset	Ask for country or state/province results.
No data for a year	Requested year is outside the active CSV	Use a supported year or ask for the latest available year.
Unexpected benchmark	Benchmark was omitted or ambiguous	State WHO, national standard, or a custom PM2.5 target explicitly.
Long answer	The request asks for many places or years	Request top 5, a shorter year range, or a specific location.

A reliable retry sequence

1. Send reset.
2. Name the geographic level and place.
3. State the metric and exact year or range.
4. State the benchmark if life loss is requested.
5. If the problem continues, share the exact question and reply with the administrator.

Administrator guide

Sign in

1. Open <https://aqli-whatsapp-chatbot.vercel.app>.
2. Sign in with the Supabase administrator email and password.
3. The application opens the conversation dashboard.

Security: Never place production passwords, service-role keys, Meta tokens, or model API keys in this manual or in chat messages. Store them only in approved environment-variable systems.

Conversation dashboard

- Search conversations by user name, phone number, or message text.
- Select a conversation to view its chronological message history.
- Unread counts reset when a conversation is opened.
- New messages and conversation updates appear through Supabase Realtime.
- Use the Knowledge base button to manage documents, CSVs, pasted text, and websites.

AI mode and Human mode

Each conversation has a mode badge in the chat header.

Mode	Behavior	Use when
AI mode	The chatbot automatically processes incoming text and sends a grounded reply.	Normal automated service
Human mode	Incoming messages are stored, but automated replies stop. An administrator can reply manually.	Escalations, sensitive questions, or manual support

1. Open the required conversation.
2. Select the AI mode/Human mode badge to switch modes.
3. In Human mode, type a response in the message field and send it.
4. Switch back to AI mode when automated replies should resume.

Knowledge-base management

Open the knowledge manager

From the conversation dashboard, select Knowledge base. The manager supports pasted text, file upload, and website import.

Supported source types

Source	Limits	Recommended use
Pasted text	Up to 2,000,000 characters	FAQs, policy notes, corrections, short methodology text
PDF	25 MB maximum	Annual reports and methodology reports
DOCX	25 MB maximum	Structured narrative documents
CSV	25 MB maximum	GADM0/GADM1 structured numeric datasets
TXT/Markdown	25 MB maximum	Plain-text references and prepared notes
Website	Single page or up to 12 indexed site pages	Public AQLI pages and selected supporting content

Upload a file

1. Enter a clear source title, including level and data year where relevant.
2. Select Upload file and choose a PDF, DOCX, CSV, TXT, or Markdown file.
3. Select Import file and wait for the chunk count confirmation.
4. Confirm that the source status is Active.
5. Run a representative WhatsApp question and verify the reference line.

CSV requirements

The current wide-year schema uses one row per geographic unit with yearly PM2.5 and life-loss columns. Common fields are shown below.

Field family	Meaning
country	Country name
name_1	State or province name

Field family	Meaning
name_2	District name; not active in the current production dataset
region, continent, iso_alpha3	Geographic classification
population	Population used for weighted averages or aggregate calculations
whostandard, natstandard	WHO and national PM2.5 benchmark fields
pm1998 ... pm2024	Annual average PM2.5 values
lpp_who_1998 ... lpp_who_2024	Life loss per person relative to WHO
lpp_nat_1998 ... lpp_nat_2024	Life loss per person relative to the national standard

Replacement behavior: Uploading a newer CSV for the same GADM level automatically deactivates older active CSV sources at that level. Verify the level classification and active status after every annual replacement.

Import a website

1. Enter a descriptive title and a public website URL.
2. Choose Single page for one URL or Site for a limited same-site crawl.
3. Select Import website. Site mode indexes up to 12 pages per import.
4. Review pages indexed, warnings, and generated chunk count.
5. Test a question whose answer appears clearly on the imported page.

Website limitation: Dynamic, login-protected, blocked, oversized, or script-only pages may not provide usable text. Import a clean PDF or pasted text when necessary.

Manage existing sources

- Deactivate a source to stop using it without deleting it.
- Activate a source to make it available to future answers.
- Reprocess a source when extraction or indexing must be rebuilt.
- Delete a source to remove it and its associated knowledge chunks.
- Keep only the intended current GADM0 and GADM1 CSVs active.

Operational troubleshooting

Problem	Administrator checks	Resolution
No WhatsApp reply	Check Vercel POST /api/webhook logs; conversation mode; Meta account status; messages webhook subscription	Restore Meta access/subscription, correct credentials, or return conversation to AI mode
Webhook verifies but replies fail	Check outbound Meta error and configured Phone Number ID	Update WhatsApp credentials/IDs in Vercel and redeploy
Wrong old data	Check active knowledge sources and CSV level classification	Upload the replacement CSV; deactivate or delete stale sources
Correct question uses old context	Review recent messages and resolved follow-up sequence	Ask user to send reset; reproduce question as standalone input
PDF/CSV upload fails	Check format, 25 MB limit, extractable text, and route logs	Clean or split the source; retry with a supported file
Website content not used	Check source status, indexed pages, warnings, and visible page text	Reprocess, import single page, or paste authoritative text
Slow response	Check grounded-answer elapsed time, model fallback, source size, and Vercel duration	Use a narrower question; review retrieval/model latency and source quality

Environment configuration categories

Production requires correctly configured environment variables in Vercel. Do not record values in operational documents.

- Supabase: project URL, publishable key, service-role key.
- WhatsApp: verify token, access token, Phone Number ID, Business Account ID.
- Model and embeddings: OpenRouter/model setting and embedding provider credentials as applicable.
- Application: APP_URL and NEXT_PUBLIC_APP_URL.

Production health check

1. Confirm the latest Vercel deployment status is Ready.
2. Verify GET /api/webhook with the configured verify token returns HTTP 200.
3. Send a real inbound WhatsApp text and confirm POST /api/webhook appears in logs.
4. Confirm a grounded answer is stored in Supabase and delivered by WhatsApp.
5. Check recent error logs for Meta send failures, timeouts, and source-processing errors.

Testing checklist and prompt library

Release acceptance checklist

- Greeting and thanks produce natural responses.
- reset clears active context but preserves message history.
- Country direct value uses GADM0 and latest year by default.
- State direct value uses GADM1.
- Top-state request without country asks for the country.
- Strict, inclusive, and equal threshold operators return correct boundaries.
- Explicit and relative period comparisons show both places.
- Country-to-state comparison cites GADM0 + GADM1 and includes a level note.
- WHO, national std, nat std, and custom target select the intended benchmark.
- Common misspellings and joined place names are handled or clarified.
- District requests state that GADM2 is unavailable.
- Every reply includes a reference and data-coverage notice.
- Human mode stops automated replies; AI mode resumes them.

Recommended test prompts

What is India's PM2.5 in 2024?

Give top five states in Nepal by PM2.5.

List countries with national std more than 30.

List states in India with PM2.5 at or below 30.

Compare India and China between 2010 and 2020.

Compare India and Uttar Pradesh for the last decade.

Compare Bihar and Uttar Pradesh life loss under the national standard.

What is national-standard life loss for Jharkhand and Uttar Pradesh for the last two years?

Show the trend for Madhesh.

Give the most polluted state in India for each year from 2010 to 2024.

What is AQLI and how is life loss calculated?

Top 10 polluted districts in India.

Glossary

Term	Definition
AQLI	Air Quality Life Index; translates long-term PM2.5 exposure into estimated life-expectancy impact
PM2.5	Fine particulate matter with aerodynamic diameter of 2.5 micrometers or smaller
GADM0	Country-level geographic dataset
GADM1	State/province-level geographic dataset
GADM2	District-level geographic dataset; not currently active
WHO benchmark	World Health Organization PM2.5 guideline field used as the default benchmark
National standard	Country-specific PM2.5 standard from the active dataset
Ilpp	Life loss per person
Period average	Average value across the selected years for each place
Annual leader	The highest or lowest place for each individual year
AI mode	Conversation mode in which automated replies are enabled
Human mode	Conversation mode in which automatic replies stop and an administrator responds
Knowledge source	An active or inactive report, website, text, or CSV available to the chatbot
Grounded answer	An answer supported by the active CSV, report, methodology document, or recent conversation context

Support and document control

Production application: <https://aqli-whatsapp-chatbot.vercel.app>

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Update this manual whenever data coverage, benchmark rules, supported message types, administrator workflows, or production platform configuration changes.